Lecture 5 – Flux and Field Lines

Parker Lamb

2020–01–21

- Drawing and interpreting electric field lines.
- Why flux – motivating Gauss' Law
- Calculating flux

Drawing electric field lines:

1. The direction fo the line indicates the direction of the field.
2. The density of the lines indicates the strength of the field.
3. Although number of total lines is arbitrary, if there is two masses, the number of lines ending on each mass must be proportional to the mass (i.e. if M1 is 2M2, M2 must have double the lines of M1).
4. Field lines come out of positive charges, go into negative.

And, some rules for drawing field lines:

1. Field lines all begin on positive charges, or infinity.
2. Field lines all end on negative charges or infinity.
3. Field lines never cross one another (if two objects, they must bend).
4. If one or more charges, number of lines beginning/ending on each charge must be proportional to the charges of each.

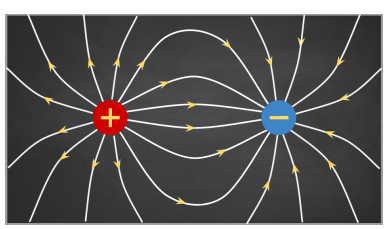
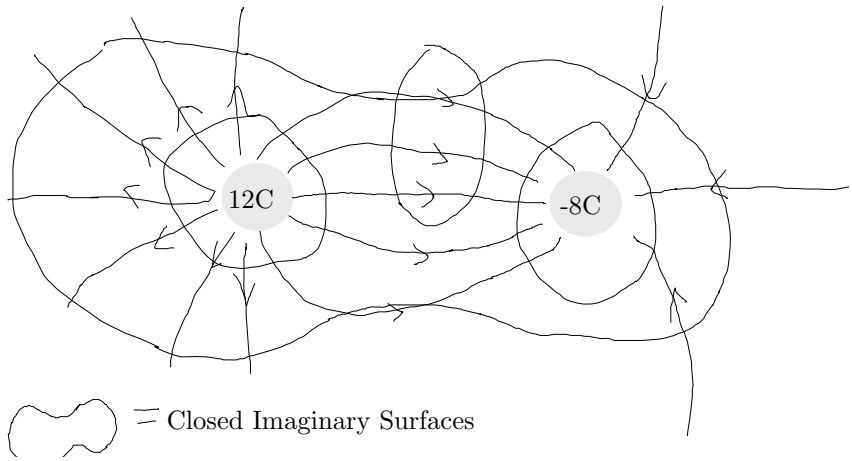


Figure 1: Field lines on dipoles.

Why flux:

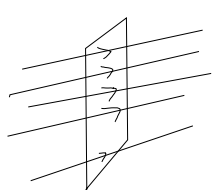
Consider the following setup, in which we have 1 field line per Coulomb charge.



Field lines leaving surface	**Charge contained inside surface**
-8	$-8C$
12	$12C$
$3-3=0$	$0C$
4	$12-8=4C$

Electric flux:

We formalize the notion of counting the number of field lines through a surface with the concept of electric flux ϕ_E, phi.



$\phi = \text{EA}$

Area of surface

$[\phi] = \frac{Nm^2}{C}$

If the E field doesn't go directly through a surface, must take account of direction.

Define area vector $\vec{A}$:

- $\vec{A}$ has magnitude A.

- $\vec{A}$ has direction perpendicular to the surface (only reasonable direction to choose).
- Take dot product into account:
  - $\vec{A} = Ax\hat{i} + Ay\hat{j} + Az\hat{k}$
  - $\vec{B} = Bx\hat{i} + By\hat{j} + Bz\hat{k}$
  - $\vec{A}.\vec{B} = AxBx + AyBy + AzBz = AB\cos\theta$

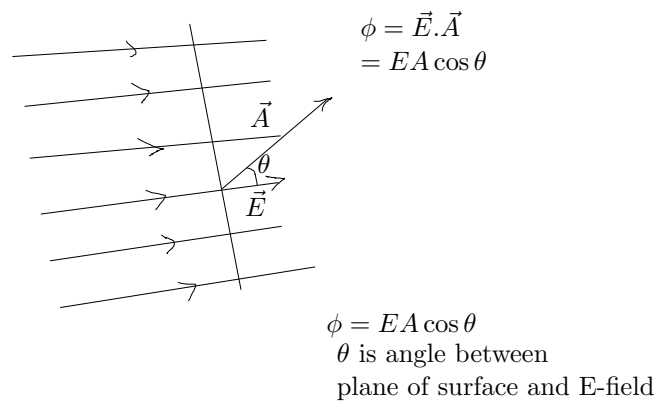


Figure 2: Area vector A